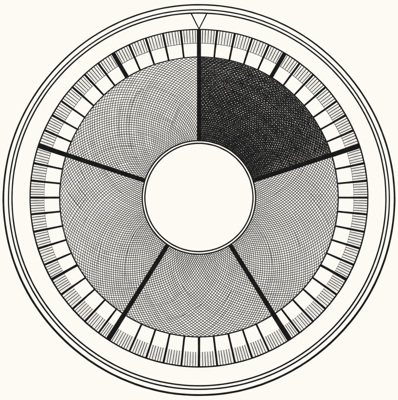


COUNTING FROM THE FIRST TICK

A Software Engineer's Attempt to Measure Time in the Universe

An introduction to ucal

Vladimir Ulogov

An introduction. The full argument is Life, the Universe, and God.

PART 1

The irritation

Read almost any account of the early universe and you will find a sentence like this one:

Recombination occurred about 380,000 years after the Big Bang.

It is a good sentence. It is doing real work. And if you stop and ask what its units are, something uncomfortable happens.

A year is the time Earth takes to go around the Sun. Not approximately — definitionally. The Julian year used in astronomy is exactly 365.25 days of exactly 86,400 seconds, and those numbers are what they are because of how fast one particular rock spins and how long it takes to circle one particular star.

So **380,000 years after the Big Bang** means 380,000 × 365.25 × 86,400 seconds, which means it is a quantity expressed in units defined by the rotation and orbit of a planet that would not exist for another nine billion years.

The number is correct. It carries a passenger.

What the complaint is, precisely

The problem is not that the units are arbitrary. All units are arbitrary. The metre was a bar in a vault, then a wavelength, then a fraction of a light-second, and none of that makes it a bad metre.

The problem is **provenance leaking into arithmetic**. When you compute with Earth years, Earth's orbital period sits inside every intermediate value. Usually that costs nothing. Occasionally it costs a rounding you did not intend, because 365.25 days is not a whole number of anything and the conversions do not close. And structurally it means the description of an event 13.8 billion years old is phrased in terms of a body that had no bearing on it.

> **NOT A CRITICISM OF COSMOLOGY**
>
> Astronomers use Julian years because they are stable, agreed and unambiguous, which is exactly what a unit should be. The complaint here is narrow, and it is aesthetic before it is technical: a quantity ought to be expressible in units that do not smuggle in a planet.

What would have to be true instead

Suppose you wanted a time system with no Earth content in its arithmetic. Four requirements fall out immediately.

A physical unit — something built from constants of nature rather than from a rotation.

A non-calendrical origin — because every calendar epoch is somebody's civil history.

A notation where writing fewer digits means saying less precisely — rather than meaning zero. At these magnitudes you are constantly quoting numbers you do not know to full precision, and a system that silently pads them is lying on your behalf.

One declared boundary with Earth — because the contact has to happen somewhere. The question is whether it happens **in the arithmetic** or at a point you can put your finger on.

That last question turned out to be the whole design. `ucal` is the answer to those four requirements, and the rest of this document is what it cost.

PART 2

What was built

TERM **Tick**

The Planck time, $t_P = \sqrt{\hbar G/c^5} \approx 5.391 \times 10^{-44}$ seconds. The atomic unit: the smallest quantity representable, and the unit in which every other quantity is counted.

The Planck time is composed from three constants — the gravitational constant, the reduced Planck constant, and the speed of light. Those bound gravitation, quantum action and causality respectively, and there is exactly one combination of them with the dimension of time.

That is the appeal. The tick is not defined by any body's motion. It does not know about Earth, or the Solar System, or matter being organised into planets at all.

THE TICK IS NOT A QUANTUM OF TIME

It is the resolution floor of an **instrument**, not a structure discovered in the world. Nothing here asserts that time comes in discrete lumps or that there is a smallest possible interval. The system cannot represent a shorter duration; that is a fact about the system.

The distinction matters because asserting otherwise would silently take a side in an argument running since Aristotle — and taking a metaphysical position as a side effect of choosing an integer type is not a respectable way to hold one.

The ladder

A tick is far too small to think in. The age of the universe is about 8×10^{60} of them, and nobody reads a 61-digit number.

So ticks are grouped into **tiers**: the powers 5^{5k}, each exactly five base-5 digits — 3125 of the tier below. The reference rung is the **beat**, 5^{60} ticks, about 46.762 milliseconds, which the specification calls the **universe second**: a unit of human-noticeable size with no Earth content whatever.

tier	name	exponent	≈
T5	deep	5^{85}	441.6 Myr
T4	drift	5^{80}	141.3 kyr
T3	span	5^{75}	45.2 yr
T2	sweep	5^{70}	5.285 d
T1	arc	5^{65}	146.1 s
T0	**beat**	5^{60}	46.762 ms
T−1	flicker	5^{55}	14.96 µs

Why base five? Because $5^5 = 3125$ is a number of exactly five base-5 digits. That is the entire reason, and it is worth saying plainly before anyone reaches the later parts of this document and starts looking for significance in the five.

What the ladder buys

A timestamp is the tick count written in base 5 and grouped in fives. Three things follow, and all three are consequences of that one fact.

Truncation is rounding. Write fewer groups and you have said the same thing less precisely. The digits you dropped are exactly the precision you gave up.

Prefix comparison is chronological comparison. Positional notation is monotone, so sorting the text sorts the times.

Writing all the digits pinpoints one tick. The coarse view and the fine view are the same integer read at different widths, so there is no accumulated conversion error between them.

What the ladder costs

Nothing on it is near anything you know. A second is 21.385 beats — not a whole number, and not close to one. An hour is 24.6 arcs. A day is 0.189 sweeps.

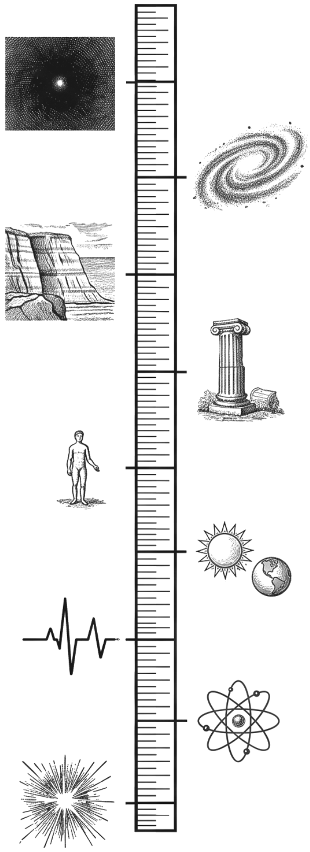
The two seconds do not divide, and the system says so on its own output: `BEAT` carries 5^{60} while `SECOND` carries only 5^{30}, so they share a common measure only at the tick. That is exactly why the tick is primitive rather than either of them.

If you want units with no planetary content, you do not get to keep the hour. The hour **is** planetary content.

The domain

The tick count is a 512-bit unsigned integer, and the ceiling is about 2.29×10^{103} years.

To scale that: proton decay, if it happens, is expected around 10^{34} years; the last black holes evaporate around 10^{100}. The domain outlasts both. The present epoch sits at 6.0×10^{-94} of the range — the counter is, for all practical purposes, still at zero.



The domain as a scale. From a burst at the Planck tick, through an atom's vibration, a heartbeat, the day, a human life, recorded history, the stratigraphic record and a galaxy's turning, to a single point of light in emptiness. Logarithmic; the present epoch sits nowhere near the top.

The width was not chosen to impress. It was chosen so that the width never has to **change**: the canonical binary encoding is 64 bytes because the domain is 512 bits, and widening it later would invalidate every stored timestamp in existence. Sixty-four bytes, paid once, against a class of migration software normally pays repeatedly.

Three refusals

Time here is **unsigned**. There is no tick −1, and a request for an instant before the datum does not return a negative number — it returns an error. There is a difference between answering a question with a negative value and refusing the question, and the system takes the second option deliberately.

That refusal is the first of three, and they are the same move at three scales:

code	rather than
`UCAL-E0020`	returning a negative tick count before the datum
`UCAL-E0031`	truncating an identifier beyond its declared range
`UCAL-E0043`	rounding an input finer than the bridge constant permits

A system that answered all three would be more convenient, and would be lying in three different ways.

No floating point

There is no floating-point value in any shipped crate of this workspace. Not in a signature, a field, a constant, an intermediate, or the rendering path that produces the human-readable output. A lint scans the source and fails the build on any float token.

The cost is concentrated in one place. Computing the age of the universe at a given redshift means evaluating an integral; in any normal library that is a call to a quadrature routine over `f64`. Here there is none, so it is certified interval quadrature over exact rationals — thousands of panels, each bounded above and below, with directed integer square roots underneath. It is slower by orders of magnitude.

What it buys is an **enclosure** rather than an estimate. The float routine returns a number and a well-informed guess about accumulated error. The interval routine returns two numbers and a proof that the true value lies between them.

> **TWO WIDTHS, NEVER MERGED**
>
> Every cosmological result carries the quadrature's error and the model's own measured uncertainty **separately**.
>
> At recombination the quadrature width is about 251 years and the parameter width about 10,900. Merged into one number the answer reads "about eleven thousand years of uncertainty", and the fact that forty-three forty-fourths of it comes from the measurement rather than the computation becomes invisible.
>
> That distinction is what tells you more computing power would buy nothing here.

The bridge

Earth enters the system through exactly one number.

the bridge constant

```
SECOND = 18 548 584 399 861 000 000 000 000 000 000 000 000 000 000
ticks
```

It is an exact integer, declared rather than computed, and it is the only place in the workspace where a foreign unit is named at all — a lint fails the build if any identifier in the core crate mentions one elsewhere.

Converting **into** absolute time is multiplication by an integer, which is exact: a whole number of seconds becomes a whole number of ticks with no rounding, ever. Converting **out** is division, which is where rounding lives — and so the rendering path is the only place a rounding mode is chosen, always explicitly.

The design does not eliminate the Earth dependency; that is not available to anyone. What it does is **localise** it. There is one constant, in one declared place, with a lint that fails the build if a second appears. The arithmetic is clean; the boundary is dirty; and the boundary is visible.

A dependency you can point at can be argued about. One dissolved into the arithmetic cannot.

PART 3

The problem with zero

A count needs somewhere to start counting from. If time is a count since some zero, the zero has to be somewhere — and the obvious place, the beginning, is not available.

Why it cannot be measured

Four reasons, and they compound.

First: exactness cannot come from measurement. The published age of the universe is 13.787 billion years, plus or minus 0.020 billion. That uncertainty, converted to ticks, is a number with fifty-eight digits — about 0.145% of the whole span from the datum to now. If the datum were defined **as** the measured age, every timestamp would inherit that wobble, and the exact integer arithmetic would be theatre over a guess.

Second: the limit is not an event. What is measured is the **age** — how long the universe has been expanding under a model — and the model is known to stop describing anything real well before you reach the limit.

Third: the extrapolation is model-dependent. The figure 13.787 is not read off an instrument; it falls out of a parameter fit. Change the model and it changes. A datum defined from it would shift with each data release.

Fourth — and this is Kant's — the question may be malformed. More on that in Part IV.

What was done instead

Three moves. Stipulate the datum. Declare the physical claim separately. Make the claim impossible to compute with.

The first two are ordinary good practice. The third is why this project turned into a book.

`BIG_BANG_CLAIM` is a value whose type has **two fields and nothing else**: no arithmetic operators of any kind, no conversion to any computable type, no method returning one. You can read its bounds, render it, print its citation. You cannot do arithmetic with it.

An absence is hard to test — you cannot assert that a method does not exist, because naming it would fail to compile. So the project uses **compile-fail tests**: programs required to fail to build, checked on every run.

tests/compile_fail/signed_window_as_operand.rs

```rust
use ucal_core::{Instant, Profile, UC1};

fn main() {
    let t: Instant<UC1> = Instant::zero();
    let claim = UC1::big_bang_claim();
    // A SignedWindow is not a Delta and must never become one.
    let _ = t.checked_add(&claim);
}
```

That is a person trying, in the most natural way available, to use the uncertainty in the age of the universe as a number. It fails to compile. The test suite passes precisely because it does.

The chain that re-executes

Declaring the claim separately would be worth little if the declaration were prose. It is data, and it runs.

ucal datum — the provenance chain

```
datum_provenance:
  input     13.787 Gyr +/- 0.020 Gyr (age_of_universe)
  citation  Planck 2018 results VI, A&A 641, A6 (2020)
  chain:
    AGE_s     = 13 787 000 000 x 31 557 600
              = 435 084 631 200 000 000 s        (exact)
    AGE_ticks = AGE_s x SECOND                   (exact)
    beats     = round_half_even(AGE_ticks / BEAT)
              = 9 304 311 741 502 590 385
    ORIGIN_OFFSET
              = beats x BEAT
  residual_rendered  -0.017190364 s
```

Every step is there: the cited input, each exact multiplication, the rounding to a whole beat, and — the part that matters most — **the residual the rounding discarded**. Seventeen milliseconds. The datum is not the published age; it is the published age rounded, and the system prints the difference rather than absorbing it.

A citation says where a number came from. This says where it came from **and what happened to it on the way in**.

PART 4

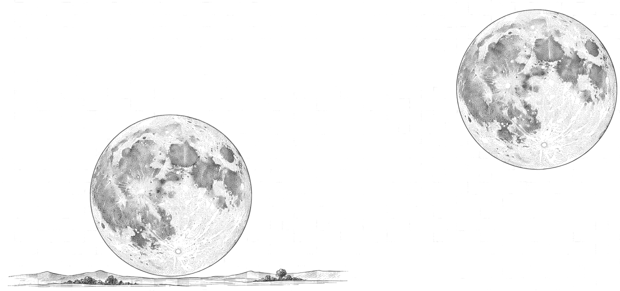
The philosophical background

This part is short on purpose. The full treatment is nine chapters of the book; what follows is the minimum needed to see that the design decisions above are not novelties.

The distinction, and why it does not stay put

There is a line between what a measurement **establishes** and what it merely **points at**. Everyone agrees it exists. Aristotle has a version, Basil of Caesarea has a version, Kant spent a large part of the first **Critique** on it — and every one of them observed that it does not stay where you put it.

Kant was the bluntest. Transcendental illusion, he says, is **natural and unavoidable**, and it **persists after diagnosis**. His image is the moon at the horizon: the astronomer knows perfectly well it is no larger there than overhead, and cannot stop seeing it larger.



The same moon, twice, drawn at the same diameter — measured on the plate the two discs differ by under one per cent. Everything else is the surrounding.

The usual remedy for this is vigilance: argue carefully, mark the boundary, and hope the next reader is paying attention. It works for as long as attention lasts.

Three borrowings

Aristotle, **Physics** IV: time is "the number of motion with respect to before and after" — not motion itself, and not a container, but the countable aspect of change. And in IV.14 he asks whether time could exist with no soul to count it, and answers carefully that there would be the substrate but not time **as number**, since number requires a numberer.

That is uncomfortable for this project rather than supportive. The datum is where a counter decided to start, and Rule Q concedes that rather than answering it.

Kant's First Antinomy supplies the fourth reason the origin cannot be measured. Thesis: the world has a beginning in time. Antithesis: it does not. Both proofs valid; both conclusions false — because both assume the world-series is a completed totality, given as a whole, about which the question has an answer waiting. If that is right, the search for the true origin is not a hard empirical problem awaiting better instruments. There is nothing of the appropriate kind to be true **of**.

Which means stipulating is not a retreat from a better method. There is no better method.

Cusanus, **De docta ignorantia**, 1440: no proportion holds between the finite and the infinite, so no increase in finite precision constitutes approach. That is this project's thesis, five and a half centuries early and stated more cleanly.

Constitutive and regulative

Kant distinguishes principles that tell you what objects **are** from principles that tell you how to go on **investigating**. Transcendental illusion is the slide from the second to the first — taking a rule for how to proceed as a description of an object.

The datum is a regulative posit: **count from here**. It does not say **here is where time began**. The physical claim is the constitutive-looking statement, and it is the one that has been rendered inert.

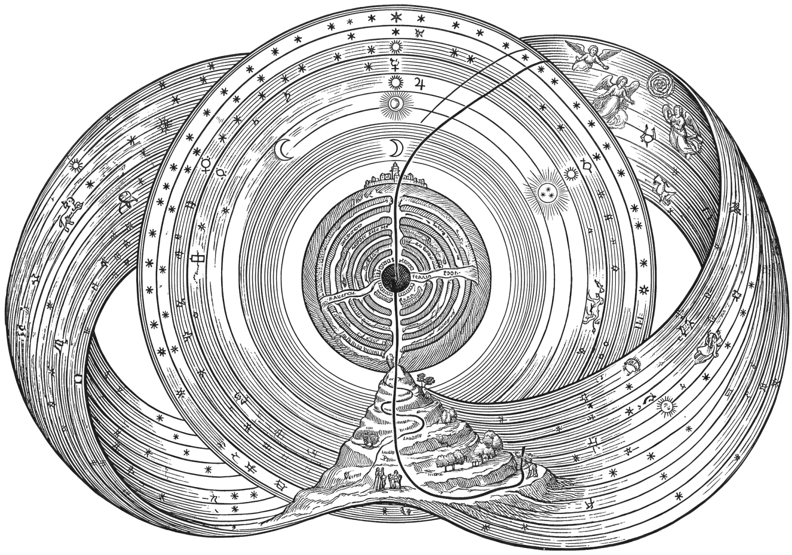
Kant had no mechanism for this. He had argument, vigilance, and the expectation that a careful reader would hold the distinction under pressure — and he says himself the illusion does not go away once diagnosed.

What this project has is not a better argument. It is the same distinction maintained by something that does not get tired, does not skim, and does not want the flattering answer.

A cautionary tale

In 1922 Pavel Florensky — mathematician, electrical engineer, Orthodox priest — published **Мнимости в геометрии**, a study of imaginary quantities in geometry.

Most of it is exactly what it says. In one part he observes that Dante's journey in the **Comedy** only **closes** — descending through the Earth, passing the centre, emerging at the Mount of Purgatory, ascending through the spheres and arriving back — if the cosmos is finite and non-orientable. That is genuine mathematical reading: it takes a structural feature of a poem seriously enough to ask what geometry the poem presupposes.



The cosmos Florensky's reading requires. The spheres twist so that the outermost rejoins the innermost; the traced line is Dante's route, in from the Empyrean, down through the infernal circles, out at the Mount of Purgatory. An illustration made for this project — not a reproduction; his book contains no plate like it.

Then, a few pages later, he considers a rigidly rotating cosmos, notes that beyond some radius the co-rotating velocity would exceed c, computes the radius at which the Lorentz factor turns imaginary — and identifies that surface with the Empyrean.

The distance between the two is exactly one move. In the first, a formal structure is used to **interpret** a text. In the second, a formal artifact is treated as **designating a place**, licensed by nothing except that the number came out at a suggestive magnitude.

He had no rule against the second move. That is not a moral observation: he had a serious apparatus and enormous facility, and nothing in his method said **here is where the mathematics stops describing and starts being read**.

This project has such a rule, and it is implemented rather than stated. It does not make the project safe — a reader looking at a 61-digit integer will still read it as a fact about being, and no type system reaches that. What the rule protects is the **arithmetic**.

PART 5

The theological background

Also short, and included for a specific reason: several of the design's decisions have precedents that are older, better named, and arrived at for entirely different purposes. Reporting them is not an argument that any tradition is correct. It is an argument that the decisions are not eccentric.

The beginning is not a member of the series

Basil of Caesarea, **Hexaemeron**, fourth century: the beginning of a road is not the road, and the beginning of a house is not the house. A beginning is not a member of the series it begins.

Applied to time: the beginning of time is not a time. Not a very short duration, not the first instant of a sequence, not a moment you could point at.

That is the datum's content, stated with no arithmetic whatsoever. Tick 0 is not the first tick of the universe; it is where counting starts, and those are different sorts of thing.

Augustine gives the operational form. Asked what God was doing before creating heaven and earth, he declines the joke and answers that there was no **then** — time is among the created things, and the question presupposes a container that does not exist. That is `UCAL-E0020` exactly: an error, not a negative number.

Interval as the mark of the created

Gregory of Nyssa, against Eunomius, makes διάστημα — interval, extension, the spread between before and after — the mark of createdness. Everything created is διαστηματικός. God is ἀδιάστατος, without interval. The gap is not one of degree along a scale but of **kind**: there is no interval there for a measure to measure.

Every quantity in this system is an interval. The tick count is the interval from the datum. A delta is an interval. A window is an interval with its uncertainty carried. The whole apparatus measures διάστημα and nothing else.

> **WHY THAT MATTERS FOR THE TITLE**
>
> An instrument for the immeasurable reads as a paradox — an instrument that cannot measure its subject is a failed instrument.
>
> Nyssa turns it into a statement about categories. The instrument reaches exactly as far as interval reaches, and adding digits does not approach what has no interval. The limit is not imprecision. It is what an interval-measure **is**.

A stipulated epoch, eleven centuries early

The Hebrew calendar's computational epoch is **molad tohu** — "the new moon of chaos" — fixed by the mnemonic BaHaRaD, and falling by rabbinic reckoning roughly a year **before** the creation it anchors.

It is computational. It is placed before the event it anchors. It is not claimed as an observation. And it is **named for its own emptiness**: tohu is the word from Genesis 1:2, formlessness, the void before ordering.

Compare the two namings. This project calls its epoch "the datum" and then spends four chapters explaining that it is stipulated rather than observed. The Hebrew calendar calls its epoch the new moon of chaos, and the explanation is in the name.

The same tradition supplies a second precedent. Time is subdivided by the **ḥelek**, 1/1080 of an hour — and 1080 was chosen because it divides cleanly, by 2, 3, 4, 5, 6, 8, 9, 10, 12 and more. That is exactly why `SECOND` is a multiple of 10^{30}: pick a constant with many factors and the subdivisions come out exact. A designer looking at a unit and asking not **how big** but **what must it divide by**.

Reliable here, not thereby necessary

Every body parameter in the system carries an epoch, a rate, and a **validity window**. Inside the window it computes confidently; outside it warns rather than extrapolating.

The epistemic content of that is: the regularity has been observed here, I will assert it here, and I decline to assert it beyond where it was observed.

Al-Ghazālī's term for the reliable-but-not-necessary regularity is ʿ**āda** — divine custom. Practical certainty within the range of custom; no claim of necessity. Whatever the author believed about causation while writing it, the posture the code takes is the one argued for in Baghdad in the eleventh century, for entirely different reasons.

AND WHERE IT CUTS THE OTHER WAY

Ibn Rushd's reply is that to deny that things have natures from which their effects follow is to deny the possibility of knowledge.

The mechanism's central promise — **choose the intercalation rule that guarantees this accuracy for this long** — presupposes that an orbital period has a nature from which its future follows. So the system computes as though Ibn Rushd were right and reports as though Ghazālī were, and it took that side silently, as an implementation consequence.

That is a metaphysical commitment the specification never declares. It was found by reading the artifact against a tradition, which is the sort of thing this part is for.

PART 6

What came out of it

A calendar with no Earth in its arithmetic turns out to be a good instrument for examining calendars that have Earth in theirs.

Every calendar in the system has the same shape: a **body**, an **anchor**, a **leap rule**, and **cycles**. Three of the four are computed from the body's own periods. The intercalation rule in particular is **derived**, by expanding the fractional day count as a continued fraction and reading off the convergents — the best rational approximations, in the precise sense that no fraction with a smaller denominator comes closer.

The Julian rule is derivable

ucal cal show earth-d — the walk

```
intercalation:
  whole_days_per_year  365
  rule                 31/128
  bound                1 local day per 10000 local years
  walked:
    1: 1/4        — 1 day slips in 128 local years
    2: 7/29       — 1 day slips in 1234 local years
    3: 8/33       — 1 day slips in 4269 local years
    4: 31/128     — 1 day slips in 400000 local years   <- chosen
    5: 752/3105   — 1 day slips in 62100000 local years
    ...
```

Convergent 1 is `1/4`: one leap day every four years. That is the Julian calendar, derived from Earth's rotation and orbit with no knowledge of Rome.

The Gregorian rule is not

Now look for `97/400` in that walk.

It is not there. Not at depth 4, not at depth 8 where the expansion terminates, **not at any depth** — and there is a check in the build whose only job is to assert its absence.

This is not an accident of where the walk stopped; it is a theorem. Convergents are the best rational approximation at their denominator, so if `97/400` were one, nothing with a smaller denominator could beat it. Two things do:

rule	denominator	1 day slips in	against 97/400
8/33	33	4,269 yr	1.32× more accurate, denominator 12.1× smaller
31/128	128	400,000 yr	**124× more accurate**, denominator 3.1× smaller
97/400	400	3,226 yr	— the Gregorian rule

This is not a criticism of the Gregorian reform, which solved the problem it was aimed at and solved it well enough to still be in use. What it means is precise: the Gregorian rule is a **declared table**, not a derivation. Its authority comes from a decision somebody made, not from arithmetic on cited parameters.

> **WHY THIS IS THE PROJECT'S BEST EVIDENCE**
>
> The specification claimed, in two published revisions, that the mechanism reproduces the Julian **and** Gregorian rules as convergents. Half of that is true.
>
> The error ran **in the direction of its author's thesis** — a machinery that rediscovers both historical calendars is a better story — and it survived two revisions because nobody checked the flattering half.
>
> A formal system built by someone with a thesis is exactly the sort of thing that can be tuned until it produces the desired result. That charge is nine hundred years old, and it admits only one kind of answer: a case where the system contradicted its builder on something already published. This is that case, and a build-time assertion now stops the correction being quietly reversed.

Two more that came free

The Metonic cycle. Earth names the Moon as its grouping satellite, and the same continued-fraction machinery applied to the year and the synodic month produces `235/19` as its sixth convergent — 235 lunar months in nineteen years. Known to Babylon, named for an Athenian, still fixing the date of Easter and the Hebrew calendar's leap years. It falls out of two numbers with nothing else supplied.

Mars has no month. Mars has two satellites; neither is anything a person would call a month. The mechanism returns nothing rather than synthesising one — because a month-shaped structure imposed on a body that has no month is Earth structure leaking through a mechanism built specifically to keep it out. Whether a satellite is "month-like" is not derivable, because **month-like** is an Earth predicate.

The absence is not a gap in the implementation. It is the implementation working.

PART 7

Where it stops

A section on capability that is longer than the section on limits is a sales document. Here are the limits.

The anchor cannot be derived

The mechanism derives units, intercalation and cycles from a body's periods. It cannot derive **phase**. Knowing exactly how long Earth takes to rotate tells you nothing about whether it is currently noon, and no amount of further tick counting will supply the missing fact.

So every calendar carries one declared, cited, interval-valued constant. The mechanism is not self-sufficient, and one component of every calendar it produces is a measurement it did not make.

Titan has no such anchor and will not be given one by invention: no published convention exists to cite. Its calendar is therefore complete in units, intercalation and cycles, and incomplete in phase — and asking it for a local date is an error rather than a guess.

Four more, briefly

A rogue planet has no year — no orbital period, no fraction to absorb, nothing to derive. Three of the four calendar components depend on a relationship between the body and something else; only rotation is the body's own.

A tidally locked body collapses two components into one — Titan's solar day is its orbit about Saturn, and its year is Saturn's about the Sun. Handled with no special case, and thinner for it.

Relativistic environments are out of scope — no time dilation, no worldline, no frame transformation. For a body deep in a gravity well this system is the wrong instrument, and it would give an answer that is wrong in a way it cannot detect.

Parameters are wrong outside their validity windows — Earth's rotation is lengthening by about 1.8 ms per century. The domain reaches 10^{103} years; the parameters are good near J2000 and degrade from there. The reach and the accuracy point in opposite directions, and the system warns rather than pretending otherwise.

What the limits have in common

Two of them are **structural** — the mechanism gives less because the body has less to give, and correct output for an unusual body is unusual output.

The other four are **epistemic**: the mechanism needs something it cannot compute. And in every one of those the response is the same — return an error or a warning, and never a default.

That consistency is the actual claim. Not that the approach works everywhere, because it does not, but that where it stops working it **says so** rather than producing a confident number nobody can audit.

A calendar that silently defaulted Titan's anchor to Earth's would work. Every function would return a value, no test would fail, and every Titanian date it produced would be wrong by an unknown amount with nothing anywhere to indicate it.

PART 8

The claim, and what is not claimed

A measuring instrument may legitimately point at what it cannot describe, provided it declares that it is only pointing — and that declaration can be enforced mechanically rather than left to the author's discipline.

The first clause is old and this document has said whose it is: Cusanus has it in 1440, Gregory of Nyssa has the reason a century earlier, and Semyon Frank reaches it independently in 1939.

The second clause is the contribution, and it amounts to one thing. A comment addresses a reader who is paying attention. A runtime check addresses a program that reaches a particular line. A **type** addresses everyone who ever writes code against the library, on every path, including the ones nobody thought about — and requires nothing of them at all.

Someone who thinks the distinction between a stipulated datum and a measured origin is pedantic sits down, writes the line that ignores it, and is stopped in under a second by something with no interest in whether they agree. They have not been persuaded. They have been refused.

What the medium costs

The argument would be cheap without this.

A program hides its premises. The Ghazālī/Ibn Rushd finding in Part V is one of three metaphysical commitments the artifact makes and never declares, all found by reading it against traditions rather than by inspecting it. An essay wears its premises; a program buries them in type signatures where they function silently.

A program's audience is smaller. The argument is fully available to people who read Rust. To everyone else it is a claim about a piece of code.

A program can be wrong with more authority. A false claim in an essay is contestable by anyone who reads it. A false claim compiled into a type is enforced against everyone downstream, and the enforcement carries no indication of whether the rule is good.

So: mechanical enforcement is a **stronger** way to maintain a distinction and a **worse** way to argue for one. That asymmetry is why there is a book as well as an artifact.

Not claimed

Not that time began — the system stipulates a datum and counts from it. It asserts nothing about whether anything started.

Not that the Big Bang happened at tick 0 — that identification is recorded as cited metadata, and made impossible to compute with.

Not that the tick is time's smallest unit — it is an instrument's resolution floor.

Not that any tradition is correct — or that any anticipated this work. There is one good algorithm for continued fractions and one obvious solution to needing an exact origin.

Not that base 5 is meaningful — $5^5 = 3125$ is five base-5 digits, and that is the whole reason.

Not that anyone should adopt it — there is one qualification about timekeeping across two or more bodies, it is a claim about coherence rather than a recommendation, and it is one page long in the book.

Usefulness is not on that list

It was, in an earlier draft, sitting sixth among the negatives — and that misfiled it. Every other item above is a claim **about the world** that the instrument declines to make. Usefulness is not that. It is a property of the artifact, and the project's position on it is structural rather than defensive.

Three things are true, in this order. The artifact is real: six crates, 381 tests, constants reproducible by two independent derivations. Its practical utility is questionable: no task you have today needs a Planck-tick count, and nothing on its ladder is near an hour. **Therefore** it is research of another kind — conducted in the medium of a working program.

The third does not rescue the second. It **depends** on it. A useful instrument would have been answerable to its users; this one is answerable only to whether it is right, and that is the whole reason the corrections in Part VI could be published rather than quietly absorbed.

What the artifact actually is

Six crates on crates.io. 381 tests on two interchangeable integer backends that accept and reject exactly the same values. A specification vendored, corrected in place, and cited by the source about a thousand times, with a build step that fails if a citation resolves to nothing. A tier table generated from the library so it cannot drift. A lint that reports every exemption it honours.

Verification found the specification wrong in fourteen places, and one further claim was raised and **withdrawn** — an error alleged in the specification that turned out to be an error in the checking oracle. The withdrawal is kept in the record, because a process that only ever finds things is indistinguishable from one that invents them.

PART 9

Where to go from here

if you want	read
the software	`cargo add ucal-core`, or `cargo install ucal` for the command line
the specification	`spec/UCAL-1.1.md` — normative, with the fourteen corrections applied in place
what the rules mean	`spec/RULES.md` — all twenty-four, with what enforces each
why it differs from the RFC	`spec/SPEC-DELTAS.md` — the record, one entry withdrawn
the full argument	**Life, the Universe, and God** — 251 pages, eight parts, nine traditions

The book is what this document is an introduction to. It contains the nine readings in full, the six samples run against real material, a chapter reserved for what those samples **failed** to establish, and the conflicts — four of which cut at the project rather than at the tradition being read.

It also ends where this document does not, on the part that the mechanism does not reach. `UCAL-E0025` stops the claim entering the arithmetic. It has no reach over what may be **read**, and there is no type signature between a number and its reader. The instrument contains the illusion; it does not cure it.

Tick zero is a stipulated reference point, conventionally identified with the FLRW $t \to 0$ limit. It is not a measurement and not an observed event.

A SOFTWARE ENGINEER'S ATTEMPT TO MEASURE TIME

— printed by `ucal datum` on every run